

AMARNATH KUMAR

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Summary

Computer Science undergraduate with hands-on experience building secure, scalable software applications. Skilled in developing authentication systems, RESTful APIs, and scalable system architectures using industry-standard web technologies. Eager to design, build, and maintain reliable, production-ready software systems.

Education

Brainware University

Bachelor of Technology in Computer Science and Engineering - AIML, CGPA: 9

2022 - 2026

Kolkata, West Bengal

Kendriya Vidyalaya Sevoke Road

Senior Secondary Examination (CBSE), Percentage: 88

2021

Siliguri, West Bengal

Experience

YBI Foundation

Nov. 2025 – Dec. 2025

Generative AI Intern

Kolkata, West Bengal

- Engineered a natural language to SQL conversion system enabling **non-technical users** to query relational databases, supporting **100+ test queries** without SQL syntax knowledge.
- Extracted and injected database schema, tables, and relationships for **10+ structured datasets**, improving SQL generation accuracy and reducing invalid queries.
- Evaluated generated SQL on sample databases and iteratively refined prompt strategies, improving query execution correctness by **25%**.

Projects

Algorithmic Assessment Engine | React.js, Node.js, Express.js

July 2025 - Oct. 2025

- Developed a full-stack coding platform offering **50+ coding questions** for practice, solution submission, and progress tracking.
- Secured **100% of user specific routes** using JWT-based authentication with persistent login and session handling.
- Implemented RESTful APIs and optimized database queries to ensure reliable request handling and consistent response times.

User Connect Social Profile System | React.js, Node.js, Express.js, MongoDB

Apr. 2025 - Jun. 2025

- Designed a public, read-only social profile system supporting discovery across **100+ user records** with controlled data visibility.
- Enforced role-based access control (RBAC) to restrict profile modifications and prevent unauthorized write operations.
- Formulated REST APIs for secure profile retrieval and username-based navigation, ensuring consistent and reliable data access.

Authentication & Authorization System | Node.js, Express.js, MongoDB, JWT

Jan. 2025 - Mar. 2025

- Built and published a **reusable Express authentication package** supporting user registration and login.
- Implemented **JWT-based authentication** with middleware-protected routes and secure password hashing.
- Applied validation and centralized error handling to ensure clean, maintainable APIs.

Skills

Languages: C++, Python, HTML, CSS, JavaScript, SQL

Frameworks & Libraries: React.js, Node.js, Express.js

Databases: MySQL, MongoDB

Developer Tools: GitHub, VS Code, Postman

Soft Skills: Problem Solving, Analytical Thinking, Team Collaboration, Time Management, Adaptability

Relevant Coursework

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|---------------------|-----------------------|---------------------------|-------------------------|
| • Data Structures | • Algorithm Analysis | • Artificial Intelligence | • Computer Networks |
| • Operating Systems | • Database Management | • Web Technologies | • Computer Architecture |

Certificates and Certifications

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|-------------------------------------|-----------------------------------|
| • Web Development Certificate | • Infosys Programming Fundamental |
| • Oracle Generative AI Professional | • Internship Certificate |